  ___ ____ ____ ____ ____ ®
 /__   /   ____/   /   ____/   **18.0**
___/   /   /___/   /   /___/   ***MP—Parallel Edition***

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Notes:
      1. Unicode is supported; see help unicode_advice.
      2. More than 2 billion observations are allowed; see help obs_advice.
      3. Maximum number of variables is set to 5,000 but can be increased; see
          help set_maxvar.

```
. 
. import delimited "/Users/chenxin/Desktop/01_DID/data/public.csv", varnames(1)
(encoding automatically selected: ISO-8859-2)
(46 vars, 410 obs)

. gen bk = (chain ==1)

. gen kfc = (chain ==2)

. gen roys = (chain ==3)

. gen nj = (state ==1)

. gen fte1 = nmgrs + empft + 0.5 * emppt
(12 missing values generated)

. label var fte1 "Wave1 full-time equivalent employment"

. gen fte2 = nmgrs2 + empft2 + 0.5 * emppt2
(14 missing values generated)

. label var fte2 "Wave2 full-time equivalent employment"

. replace fte2 = 0 if status2 ==4
(1 real change made)

. gen demp = fte2 - fte1
(25 missing values generated)

. label var demp "Change in FTE employment (fte2-fte1)"
```

```
14 . gen gap = 0

15 . replace gap = (5.05 - wage_st)/wage_st  if state ==1 & wage_st <5.05
   (286 real changes made)

16 . label var gap "Wage gap: proportional required wage increase"

17 . gen pmeal = psoda + pfry + pentree
   (23 missing values generated)

18 . gen pmeal2 = psoda2 + pfry2 + pentree2
   (34 missing values generated)

19 . gen dln_pmeal = ln(pmeal2) - ln(pmeal)
   (54 missing values generated)

20 . label var dln_pmeal "Log change of full-meal price"

21 . preserve

22 . keep if demp ~=. & wage_st ~=. & wage_st2 ~=.
   (59 observations deleted)

23 . count
     351

24 . reg demp nj, robust

   Linear regression                               Number of obs     =        351
                                                   F(1, 349)         =       2.46
                                                   Prob > F          =     0.1174
                                                   R-squared         =     0.0104
                                                   Root MSE          =     8.7146

------------------------------------------------------------------------------
             |               Robust
        demp |  Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
          nj |   2.276858   1.450472     1.57   0.117    -.5759083    5.129624
       _cons |  -1.878788   1.369786    -1.37   0.171    -4.572862    .8152867
------------------------------------------------------------------------------

25 . reg demp nj bk kfc roys co_owned, robust

   Linear regression                               Number of obs     =        351
                                                   F(5, 345)         =       1.65
                                                   Prob > F          =     0.1463
                                                   R-squared         =     0.0204
                                                   Root MSE          =     8.7205

------------------------------------------------------------------------------
             |               Robust
        demp |  Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
```

nj	2.281507	1.45291	1.57	0.117	-.5761694	5.139184
bk	.7566441	1.636293	0.46	0.644	-2.461721	3.975009
kfc	.9911949	1.5663	0.63	0.527	-2.089505	4.071895
roys	-1.328043	1.608087	-0.83	0.409	-4.490931	1.834846
co_owned	.3729144	.8967168	0.42	0.678	-1.390806	2.136634
_cons	-2.206675	1.830356	-1.21	0.229	-5.806737	1.393386

26 . reg demp gap, robust

Linear regression

Number of obs	351
F(1, 349)	7.69
Prob > F	0.0059
R-squared	0.0220
Root MSE	8.6634

demp	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	17.05158	6.149891	2.77	0.006	4.956066	29.14709
_cons	-1.477734	.7725831	-1.91	0.057	-2.997239	.0417704

27 . reg demp gap bk kfc roys co_owned, robust

Linear regression

Number of obs	351
F(5, 345)	2.44
Prob > F	0.0345
R-squared	0.0294
Root MSE	8.6802

demp	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	16.36307	6.541488	2.50	0.013	3.496852	29.22929
bk	.1705731	1.696863	0.10	0.920	-3.166926	3.508072
kfc	.5643347	1.604376	0.35	0.725	-2.591255	3.719925
roys	-1.558957	1.617539	-0.96	0.336	-4.740436	1.622522
co_owned	.5103669	.9074505	0.56	0.574	-1.274465	2.295198
_cons	-1.402613	1.452848	-0.97	0.335	-4.260167	1.454941

28 . reg demp gap bk kfc roys co_owned southj centralj northj pa1 pa2, robust
note: **pa1** omitted because of collinearity.

Linear regression

Number of obs	351
F(9, 341)	1.95
Prob > F	0.0442
R-squared	0.0430
Root MSE	8.6695

demp	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	13.87905	7.061524	1.97	0.050	-.0105832	27.76868
bk	-.1166013	1.640058	-0.07	0.943	-3.342506	3.109303
kfc	.2801595	1.61505	0.17	0.862	-2.896556	3.456875
roys	-2.029289	1.622368	-1.25	0.212	-5.220397	1.161819
co_owned	.2192472	.9413176	0.23	0.816	-1.632273	2.070767
southj	3.508251	2.013822	1.74	0.082	-.4528263	7.469328
centralj	2.154794	2.348667	0.92	0.360	-2.464904	6.774493
northj	3.481748	1.998569	1.74	0.082	-.449327	7.412823
pa1	0	(omitted)				
pa2	4.14951	2.73044	1.52	0.130	-1.221115	9.520136
_cons	-3.875872	2.078513	-1.86	0.063	-7.964193	.2124491

```
29 .
30 . restore

31 . preserve

32 . gen fte2_rob = fte2
   (13 missing values generated)

33 . replace fte2_rob = 0 if status2 ==3
   (0 real changes made)

34 . gen demp_rob = fte2_rob - fte1
   (25 missing values generated)

35 . keep if demp_rob ~=. & wage_st ~=. & wage_st2 ~=.
   (59 observations deleted)

36 . reg demp_rob gap bk kfc roys co_owned, robust

   Linear regression                               Number of obs     =        351
                                                   F(5, 345)         =       2.44
                                                   Prob > F          =     0.0345
                                                   R-squared         =     0.0294
                                                   Root MSE          =     8.6802
```

demp_rob	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	16.36307	6.541488	2.50	0.013	3.496852	29.22929
bk	.1705731	1.696863	0.10	0.920	-3.166926	3.508072
kfc	.5643347	1.604376	0.35	0.725	-2.591255	3.719925
roys	-1.558957	1.617539	-0.96	0.336	-4.740436	1.622522
co_owned	.5103669	.9074505	0.56	0.574	-1.274465	2.295198
_cons	-1.402613	1.452848	-0.97	0.335	-4.260167	1.454941

```
37 . restore
```

```
38 . preserve

39 . keep if dln_pmeal ~=. & gap~=. & bk~=. &kfc~=. & roys~=. & co_owned~=.
   (54 observations deleted)

40 . count
     356

41 . reg dln_pmeal nj bk kfc roys co_owned, robust

   Linear regression                               Number of obs     =        356
                                                   F(5, 350)         =       2.33
                                                   Prob > F          =     0.0422
                                                   R-squared         =     0.0625
                                                   Root MSE          =      .1002

------------------------------------------------------------------------------
             |               Robust
   dln_pmeal |  Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
          nj |   .0315689   .0129919     2.43   0.016     .0060168     .057121
          bk |  -.0640087   .0265011    -2.42   0.016      -.11613   -.0118873
         kfc |  -.0680462   .0277743    -2.45   0.015    -.1226717   -.0134207
        roys |  -.0655214   .0308164    -2.13   0.034    -.1261301   -.0049128
   co_owned |   .0018764   .0132572     0.14   0.888    -.0241974    .0279501
       _cons |    .044266    .027383     1.62   0.107      -.00959     .098122
------------------------------------------------------------------------------

42 . reg dln_pmeal gap bk kfc roys co_owned southj centralj northj pa1 pa2, robust
   note: pa2 omitted because of collinearity.

   Linear regression                               Number of obs     =        356
                                                   F(9, 346)         =       2.46
                                                   Prob > F          =     0.0100
                                                   R-squared         =     0.0742
                                                   Root MSE          =     .10015

------------------------------------------------------------------------------
             |               Robust
   dln_pmeal |  Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         gap |   .0834646    .070525     1.18   0.237    -.0552472    .2221763
          bk |  -.0692218   .0257875    -2.68   0.008    -.1199417   -.0185018
         kfc |  -.0708994   .0271982    -2.61   0.010    -.1243941   -.0174047
        roys |   -.070445   .0307961    -2.29   0.023    -.1310162   -.0098738
   co_owned |  -.0000326   .0133132    -0.00   0.998    -.0262177    .0261525
     southj |   .0119662   .0183191     0.65   0.514    -.0240646     .047997
   centralj |  -.0061224   .0225821    -0.27   0.786    -.0505379    .0382931
     northj |   .0191468   .0155534     1.23   0.219    -.0114443    .0497378
        pa1 |  -.0247439   .0272327    -0.91   0.364    -.0783064    .0288186
        pa2 |           0  (omitted)
       _cons |   .0593612    .028608     2.07   0.039     .0030938    .1156286
------------------------------------------------------------------------------
```

```
43 . restore

44 .
45 . summarize fte1 fte2 demp nj gap wage_st wage_st2 dln_pmeal
```

Variable	Obs	Mean	Std. dev.	Min	Max
fte1	398	20.99887	9.749805	5	85
fte2	397	21.00126	9.144222	0	60.5
demp	385	-.117013	9.056784	-41.5	34
nj	410	.8073171	.394888	0	1
gap	410	.0803588	.0768139	0	.1882353
wage_st	390	4.615641	.3470151	4.25	5.75
wage_st2	389	4.996273	.2531905	4.25	6.25
dln_pmeal	356	.0145711	.1027562	-.3955148	.4739851

```
46 . bysort nj: summarize fte1 fte2 demp wage_st wage_st2

-> nj = 0
```

Variable	Obs	Mean	Std. dev.	Min	Max
fte1	77	23.33117	11.85628	7.5	70.5
fte2	77	21.16558	8.276732	0	43.5
demp	75	-2.283333	10.85363	-41.5	22.75
wage_st	76	4.630132	.3516873	4.25	5.5
wage_st2	71	4.617465	.3574781	4.25	6.25

```
-> nj = 1
```

Variable	Obs	Mean	Std. dev.	Min	Max
fte1	321	20.43941	9.106239	5	85
fte2	320	20.96172	9.352609	0	60.5
demp	310	.4070968	8.503438	-34	34
wage_st	314	4.612134	.3463505	4.25	5.75
wage_st2	318	5.080849	.1045445	5	5.75

```
47 .
48 . *生成NJ内部工资分组

49 . gen wage_group = .
   (410 missing values generated)

50 . replace wage_group =1 if nj==1 & wage_st==4.25
   (101 real changes made)

51 . replace wage_group =2 if nj==1 & wage_st>4.25 & wage_st<5.00
   (140 real changes made)
```

```
52 . replace wage_group =3 if nj==1 & wage_st>=5.00
   (90 real changes made)

53 . replace wage_group =0 if nj==0 // PA组
   0/ invalid name
   r(198);

54 .
55 . *计算每组fte1 fte2 demp的均值

56 . tabstat fte1 fte2 demp, by(wage_group) statistics(n mean sd)

   Summary statistics: N, Mean, SD
   Group variable: wage_group
```

wage_group	fte1	fte2	demp
1	97	98	94
	19.5567	20.87755	1.204787
	7.631444	9.957917	7.974334
2	137	135	132
	20.08066	20.95556	.7098485
	9.846855	8.779368	7.884145
3	87	87	84
	21.98851	21.06609	-.9613095
	9.311151	9.617576	9.852718
Total	321	320	310
	20.43941	20.96172	.4070968
	9.106239	9.352609	8.503438

```
57 .
58 . replace wage_group =0 if nj==0
   (79 real changes made)

59 . tabstat fte1 fte2 demp, by(wage_group) statistics(n mean sd)

   Summary statistics: N, Mean, SD
   Group variable: wage_group
```

wage_group	fte1	fte2	demp
0	77	77	75
	23.33117	21.16558	-2.283333
	11.85628	8.276732	10.85363
1	97	98	94
	19.5567	20.87755	1.204787
	7.631444	9.957917	7.974334

2	137	135	132
	20.08066	20.95556	.7098485
	9.846855	8.779368	7.884145
3	87	87	84
	21.98851	21.06609	-.9613095
	9.311151	9.617576	9.852718
Total	398	397	385
	20.99887	21.00126	-.117013
	9.749805	9.144222	9.056784

```
60 . preserve

61 . keep if demp ~=. & wage_st ~=. & wage_st2 ~=.
   (59 observations deleted)

62 .
63 . *Table4

64 . reg demp nj, robust

   Linear regression                               Number of obs     =        351
                                                   F(1, 349)         =       2.46
                                                   Prob > F          =     0.1174
                                                   R-squared         =     0.0104
                                                   Root MSE          =     8.7146
```

demp	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
nj	2.276858	1.450472	1.57	0.117	-.5759083	5.129624
_cons	-1.878788	1.369786	-1.37	0.171	-4.572862	.8152867

```
65 . reg demp nj bk kfc roys co_owned, robust

   Linear regression                               Number of obs     =        351
                                                   F(5, 345)         =       1.65
                                                   Prob > F          =     0.1463
                                                   R-squared         =     0.0204
                                                   Root MSE          =     8.7205
```

demp	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
nj	2.281507	1.45291	1.57	0.117	-.5761694	5.139184
bk	.7566441	1.636293	0.46	0.644	-2.461721	3.975009
kfc	.9911949	1.5663	0.63	0.527	-2.089505	4.071895
roys	-1.328043	1.608087	-0.83	0.409	-4.490931	1.834846
co_owned	.3729144	.8967168	0.42	0.678	-1.390806	2.136634

_cons	**-2.206675**	**1.830356**	**-1.21**	**0.229**	**-5.806737**	**1.393386**

66 . reg demp gap, robust

Linear regression

Number of obs	**351**
F(1, 349)	**7.69**
Prob > F	**0.0059**
R-squared	**0.0220**
Root MSE	**8.6634**

demp	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	**17.05158**	**6.149891**	**2.77**	**0.006**	**4.956066**	**29.14709**
_cons	**-1.477734**	**.7725831**	**-1.91**	**0.057**	**-2.997239**	**.0417704**

67 . reg demp gap bk kfc roys co_owned, robust

Linear regression

Number of obs	**351**
F(5, 345)	**2.44**
Prob > F	**0.0345**
R-squared	**0.0294**
Root MSE	**8.6802**

demp	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	**16.36307**	**6.541488**	**2.50**	**0.013**	**3.496852**	**29.22929**
bk	**.1705731**	**1.696863**	**0.10**	**0.920**	**-3.166926**	**3.508072**
kfc	**.5643347**	**1.604376**	**0.35**	**0.725**	**-2.591255**	**3.719925**
roys	**-1.558957**	**1.617539**	**-0.96**	**0.336**	**-4.740436**	**1.622522**
co_owned	**.5103669**	**.9074505**	**0.56**	**0.574**	**-1.274465**	**2.295198**
_cons	**-1.402613**	**1.452848**	**-0.97**	**0.335**	**-4.260167**	**1.454941**

68 . reg demp gap bk kfc roys co_owned southj centralj northj pa1 pa2, robust
note: **pa1** omitted because of collinearity.

Linear regression

Number of obs	**351**
F(9, 341)	**1.95**
Prob > F	**0.0442**
R-squared	**0.0430**
Root MSE	**8.6695**

demp	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	**13.87905**	**7.061524**	**1.97**	**0.050**	**-.0105832**	**27.76868**
bk	**-.1166013**	**1.640058**	**-0.07**	**0.943**	**-3.342506**	**3.109303**

kfc	.2801595	1.61505	0.17	0.862	-2.896556	3.456875
roys	-2.029289	1.622368	-1.25	0.212	-5.220397	1.161819
co_owned	.2192472	.9413176	0.23	0.816	-1.632273	2.070767
southj	3.508251	2.013822	1.74	0.082	-.4528263	7.469328
centralj	2.154794	2.348667	0.92	0.360	-2.464904	6.774493
northj	3.481748	1.998569	1.74	0.082	-.449327	7.412823
pa1	0	(omitted)				
pa2	4.14951	2.73044	1.52	0.130	-1.221115	9.520136
_cons	-3.875872	2.078513	-1.86	0.063	-7.964193	.2124491

```
69 .
70 . restore

71 .
72 . preserve

73 . keep if demp ~=. & wage_st ~=. & wage_st2 ~=.
   (59 observations deleted)

74 .
75 . gen fte2_r2 = fte2

76 . replace fte2_r2 =0 if status2 ==3
   (0 real changes made)

77 . gen demp_r2 = fte2_r2 - fte1

78 . reg demp_r2 gap bk kfc roys co_owned, robust

   Linear regression                               Number of obs     =        351
                                                   F(5, 345)         =       2.44
                                                   Prob > F          =     0.0345
                                                   R-squared         =     0.0294
                                                   Root MSE          =     8.6802
```

demp_r2	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	16.36307	6.541488	2.50	0.013	3.496852	29.22929
bk	.1705731	1.696863	0.10	0.920	-3.166926	3.508072
kfc	.5643347	1.604376	0.35	0.725	-2.591255	3.719925
roys	-1.558957	1.617539	-0.96	0.336	-4.740436	1.622522
co_owned	.5103669	.9074505	0.56	0.574	-1.274465	2.295198
_cons	-1.402613	1.452848	-0.97	0.335	-4.260167	1.454941

```
79 .
80 . gen fte1_nomgr = empft +0.5*emppt

81 . gen fte2_nomgr = empft2 +0.5*emppt2

82 . replace fte2_nomgr =0 if status2==4
```

```
   (0 real changes made)

83 . gen demp_nomgr = fte2_nomgr - fte1_nomgr

84 . reg demp_nomgr gap bk kfc roys co_owned, robust

   Linear regression                               Number of obs     =        351
                                                   F(5, 345)         =       2.38
                                                   Prob > F          =     0.0382
                                                   R-squared         =     0.0286
                                                   Root MSE          =     8.5193

------------------------------------------------------------------------------
             |               Robust
  demp_nomgr |  Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         gap |    15.78565   6.434896     2.45   0.015     3.129086    28.44222
          bk |    .2195208   1.664824     0.13   0.895    -3.054962    3.494004
         kfc |    .6802062   1.543928     0.44   0.660    -2.356491    3.716903
        roys |   -1.396248   1.576169    -0.89   0.376    -4.496359    1.703862
   co_owned |    .4683181    .885664     0.53   0.597    -1.273662    2.210299
       _cons |   -1.539669   1.409547    -1.09   0.275    -4.312056    1.232718
------------------------------------------------------------------------------

85 .
86 . gen fte1_w4 = nmgrs + empft + 0.4*emppt

87 . gen fte2_w4 = nmgrs2 + empft2 +0.4*emppt2

88 . replace fte2_w4=0 if status2==4
   (0 real changes made)

89 . gen demp_w4 = fte2_w4-fte1_w4

90 . reg demp_w4 gap bk kfc roys co_owned, robust

   Linear regression                               Number of obs     =        351
                                                   F(5, 345)         =       2.53
                                                   Prob > F          =     0.0287
                                                   R-squared         =     0.0305
                                                   Root MSE          =     8.7518

------------------------------------------------------------------------------
             |               Robust
     demp_w4 |  Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         gap |    16.58176   6.556878     2.53   0.012      3.68527    29.47824
          bk |    .5436982   1.729618     0.31   0.753    -2.858225    3.945621
         kfc |    .8511961   1.651547     0.52   0.607    -2.397171    4.099563
        roys |   -1.313564   1.661109    -0.79   0.430    -4.580741    1.953612
   co_owned |    .6982864   .9161634     0.76   0.446    -1.103682    2.500255
       _cons |   -1.742924   1.495139    -1.17   0.245    -4.683658     1.19781
------------------------------------------------------------------------------
```

```
91 .
92 . gen fte1_w6 = nmgrs + empft + 0.6*emppt

93 . gen fte2_w6 = nmgrs2 + empft2 +0.6*emppt2

94 . replace fte2_w6=0 if status2==4
   (0 real changes made)

95 . gen demp_w6 = fte2_w6-fte1_w6

96 . reg demp_w6 gap bk kfc roys co_owned, robust

   Linear regression                               Number of obs     =        351
                                                   F(5, 345)         =       2.32
                                                   Prob > F          =     0.0428
                                                   R-squared         =     0.0281
                                                   Root MSE          =      8.724

------------------------------------------------------------------------------
             |               Robust
     demp_w6 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         gap |   16.14438   6.612595     2.44   0.015     3.138306    29.15045
          bk |  -.2025521    1.68675    -0.12   0.904     -3.52016    3.115056
         kfc |   .2774736    1.57782     0.18   0.861    -2.825884    3.380831
        roys |  -1.804349   1.596919    -1.13   0.259    -4.945272    1.336574
   co_owned |   .3224474   .9149994     0.35   0.725    -1.477232    2.122127
       _cons |  -1.062303   1.431411    -0.74   0.459    -3.877693    1.753087
------------------------------------------------------------------------------

97 .
98 . reg demp gap bk kfc roys co_owned if ncalls <=2, robust

   Linear regression                               Number of obs     =        289
                                                   F(5, 283)         =       1.51
                                                   Prob > F          =     0.1882
                                                   R-squared         =     0.0223
                                                   Root MSE          =     9.0942

------------------------------------------------------------------------------
             |               Robust
        demp | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         gap |   15.20406   7.451796     2.04   0.042     .5360812    29.87204
          bk |   .5669525   2.026482     0.28   0.780    -3.421938    4.555843
         kfc |   .9745721   1.940074     0.50   0.616    -2.844235    4.793379
        roys |  -.6216868   1.927394    -0.32   0.747    -4.415535    3.172162
   co_owned |   .3096648   1.026155     0.30   0.763    -1.710201    2.329531
       _cons |  -1.729017   1.756672    -0.98   0.326    -5.186819    1.728784
------------------------------------------------------------------------------

99 .
100 . reg demp gap bk kfc roys co_owned [aweight=fte1], robust
   (sum of wgt is 7,457.79999923706)
```

```
Linear regression                               Number of obs     =        351
                                                F(5, 345)         =       2.89
                                                Prob > F          =     0.0142
                                                R-squared         =     0.0573
                                                Root MSE          =     10.478

------------------------------------------------------------------------------
             |               Robust
        demp |Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         gap |   31.90903   10.09462     3.16   0.002     12.05429    51.76378
          bk |  -1.258513   2.729152    -0.46   0.645    -6.626383    4.109358
         kfc |   .7431318   2.504407     0.30   0.767    -4.182696     5.66896
        roys |  -1.728685   2.499055    -0.69   0.490    -6.643987    3.186616
   co_owned |   .9311282   1.269894     0.73   0.464     -1.56658    3.428836
       _cons |  -4.264698   2.292517    -1.86   0.064    -8.773768    .2443714
------------------------------------------------------------------------------

101 .
102 . preserve
already preserved
r(621);

103 . keep if nj==0
(285 observations deleted)

104 . gen fake_gap = (5.05 - wage_st)/wage_st

105 . reg demp fake_gap bk kfc roys co_owned, robust

Linear regression                               Number of obs     =         66
                                                F(5, 60)          =       3.47
                                                Prob > F          =     0.0081
                                                R-squared         =     0.0370
                                                Root MSE          =     11.421

------------------------------------------------------------------------------
             |               Robust
        demp |Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    fake_gap |  -3.240502   22.39365    -0.14   0.885    -48.03448    41.55348
          bk |  -.5494061   4.178212    -0.13   0.896    -8.907074    7.808262
         kfc |   6.377866   3.163369     2.02   0.048     .0501848    12.70555
        roys |   .9851063   3.921605     0.25   0.803    -6.859271    8.829483
   co_owned |  -2.488415   3.568191    -0.70   0.488     -9.62586    4.649031
       _cons |  -1.617904   4.807863    -0.34   0.738    -11.23506    7.999253
------------------------------------------------------------------------------

106 . restore

107 .
108 . restore
nothing to restore
```

```
    r(622);

109 .
110 . preserve

111 . keep if dln_pmeal ~=. & gap~=. & bk~=. &kfc~=. & roys~=. & co_owned~=.
    (54 observations deleted)

112 . count
      356

113 . reg dln_pmeal nj, robust

    Linear regression                               Number of obs     =        356
                                                    F(1, 354)         =       4.96
                                                    Prob > F          =     0.0265
                                                    R-squared         =     0.0122
                                                    Root MSE          =     .10227

------------------------------------------------------------------------------
             |               Robust
  dln_pmeal  |  Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
          nj |   .0288594   .0129554     2.23   0.027     .0033802    .0543385
       _cons |  -.0087758   .0114109    -0.77   0.442    -.0312175    .0136659
------------------------------------------------------------------------------

114 . reg dln_pmeal nj bk kfc roys co_owned, robust

    Linear regression                               Number of obs     =        356
                                                    F(5, 350)         =       2.33
                                                    Prob > F          =     0.0422
                                                    R-squared         =     0.0625
                                                    Root MSE          =      .1002

------------------------------------------------------------------------------
             |               Robust
  dln_pmeal  |  Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
          nj |   .0315689   .0129919     2.43   0.016     .0060168     .057121
          bk |  -.0640087   .0265011    -2.42   0.016      -.11613   -.0118873
         kfc |  -.0680462   .0277743    -2.45   0.015    -.1226717   -.0134207
        roys |  -.0655214   .0308164    -2.13   0.034    -.1261301   -.0049128
    co_owned |   .0018764   .0132572     0.14   0.888    -.0241974    .0279501
       _cons |    .044266    .027383     1.62   0.107      -.00959     .098122
------------------------------------------------------------------------------

115 . reg dln_pmeal gap, robust

    Linear regression                               Number of obs     =        356
                                                    F(1, 354)         =       1.87
                                                    Prob > F          =     0.1719
                                                    R-squared         =     0.0041
                                                    Root MSE          =     .10269
```

dln_pmeal	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	.0857754	.0626548	1.37	0.172	-.0374472	.2089979
_cons	.0075877	.0085193	0.89	0.374	-.0091672	.0243425

```
116 . reg dln_pmeal gap bk kfc roys co_owned, robust

Linear regression                               Number of obs     =        356
                                                F(5, 350)         =       3.64
                                                Prob > F          =     0.0032
                                                R-squared         =     0.0594
                                                Root MSE          =     .10037
```

dln_pmeal	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	.1466354	.0544056	2.70	0.007	.0396325	.2536384
bk	-.0695723	.0261646	-2.66	0.008	-.1210319	-.0181126
kfc	-.0722437	.0275398	-2.62	0.009	-.126408	-.0180793
roys	-.0675917	.030745	-2.20	0.029	-.1280599	-.0071235
co_owned	.0019237	.0136894	0.14	0.888	-.0250002	.0288476
_cons	.061525	.0272451	2.26	0.025	.0079402	.1151097

```
117 . reg dln_pmeal gap bk kfc roys co_owned southj centralj northj pa1 pa2, robust
note: pa2 omitted because of collinearity.

Linear regression                               Number of obs     =        356
                                                F(9, 346)         =       2.46
                                                Prob > F          =     0.0100
                                                R-squared         =     0.0742
                                                Root MSE          =     .10015
```

dln_pmeal	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	.0834646	.070525	1.18	0.237	-.0552472	.2221763
bk	-.0692218	.0257875	-2.68	0.008	-.1199417	-.0185018
kfc	-.0708994	.0271982	-2.61	0.010	-.1243941	-.0174047
roys	-.070445	.0307961	-2.29	0.023	-.1310162	-.0098738
co_owned	-.0000326	.0133132	-0.00	0.998	-.0262177	.0261525
southj	.0119662	.0183191	0.65	0.514	-.0240646	.047997
centralj	-.0061224	.0225821	-0.27	0.786	-.0505379	.0382931
northj	.0191468	.0155534	1.23	0.219	-.0114443	.0497378
pa1	-.0247439	.0272327	-0.91	0.364	-.0783064	.0288186
pa2	0	(omitted)				
_cons	.0593612	.028608	2.07	0.039	.0030938	.1156286

```
118 . restore

119 .
120 . ssc install estout
    checking estout consistency and verifying not already installed...

    the following files already exist and are different:
        /Users/chenxin/Library/Application Support/Stata/ado/plus/e/estout.ado
        /Users/chenxin/Library/Application Support/Stata/ado/plus/e/estout.hlp
        /Users/chenxin/Library/Application Support/Stata/ado/plus/e/esttab.ado
        /Users/chenxin/Library/Application Support/Stata/ado/plus/e/esttab.hlp

    no files installed or copied
    (no action taken)
    r(602);

121 . preserve

122 . keep if demp ~=. & wage_st ~=. & wage_st2 ~=.
    (59 observations deleted)

123 . eststo clear

124 . eststo: reg demp nj, robust

    Linear regression                               Number of obs     =        351
                                                    F(1, 349)         =       2.46
                                                    Prob > F          =     0.1174
                                                    R-squared         =     0.0104
                                                    Root MSE          =     8.7146

    ------------------------------------------------------------------------------
                 |               Robust
           demp |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
    -------------+----------------------------------------------------------------
              nj |   2.276858   1.450472     1.57   0.117    -.5759083    5.129624
           _cons |  -1.878788   1.369786    -1.37   0.171    -4.572862    .8152867
    ------------------------------------------------------------------------------

    (est1 stored)

125 . eststo: reg demp nj bk kfc roys co_owned, robust

    Linear regression                               Number of obs     =        351
                                                    F(5, 345)         =       1.65
                                                    Prob > F          =     0.1463
                                                    R-squared         =     0.0204
                                                    Root MSE          =     8.7205

    ------------------------------------------------------------------------------
                 |               Robust
           demp |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
    -------------+----------------------------------------------------------------
              nj |   2.281507    1.45291     1.57   0.117    -.5761694    5.139184
```

bk	.7566441	1.636293	0.46	0.644	-2.461721	3.975009
kfc	.9911949	1.5663	0.63	0.527	-2.089505	4.071895
roys	-1.328043	1.608087	-0.83	0.409	-4.490931	1.834846
co_owned	.3729144	.8967168	0.42	0.678	-1.390806	2.136634
_cons	-2.206675	1.830356	-1.21	0.229	-5.806737	1.393386

(**est2** stored)

126 . eststo: reg demp gap, robust

Linear regression

Number of obs	= 351
F(1, 349)	= 7.69
Prob > F	= 0.0059
R-squared	= 0.0220
Root MSE	= 8.6634

demp	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	17.05158	6.149891	2.77	0.006	4.956066	29.14709
_cons	-1.477734	.7725831	-1.91	0.057	-2.997239	.0417704

(**est3** stored)

127 . eststo: reg demp gap bk kfc roys co_owned, robust

Linear regression

Number of obs	= 351
F(5, 345)	= 2.44
Prob > F	= 0.0345
R-squared	= 0.0294
Root MSE	= 8.6802

demp	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	16.36307	6.541488	2.50	0.013	3.496852	29.22929
bk	.1705731	1.696863	0.10	0.920	-3.166926	3.508072
kfc	.5643347	1.604376	0.35	0.725	-2.591255	3.719925
roys	-1.558957	1.617539	-0.96	0.336	-4.740436	1.622522
co_owned	.5103669	.9074505	0.56	0.574	-1.274465	2.295198
_cons	-1.402613	1.452848	-0.97	0.335	-4.260167	1.454941

(**est4** stored)

128 . eststo: reg demp gap bk kfc roys co_owned southj centralj northj pa1 pa2, robust
note: **pa1** omitted because of collinearity.

Linear regression

Number of obs	= 351
F(9, 341)	= 1.95
Prob > F	= 0.0442
R-squared	= 0.0430
Root MSE	= 8.6695

demp	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	13.87905	7.061524	1.97	0.050	-.0105832	27.76868
bk	-.1166013	1.640058	-0.07	0.943	-3.342506	3.109303
kfc	.2801595	1.61505	0.17	0.862	-2.896556	3.456875
roys	-2.029289	1.622368	-1.25	0.212	-5.220397	1.161819
co_owned	.2192472	.9413176	0.23	0.816	-1.632273	2.070767
southj	3.508251	2.013822	1.74	0.082	-.4528263	7.469328
centralj	2.154794	2.348667	0.92	0.360	-2.464904	6.774493
northj	3.481748	1.998569	1.74	0.082	-.449327	7.412823
pa1	0	(omitted)				
pa2	4.14951	2.73044	1.52	0.130	-1.221115	9.520136
_cons	-3.875872	2.078513	-1.86	0.063	-7.964193	.2124491

(**est5** stored)

```
129 . esttab using table4.rtf, replace se star(* 0.1 **0.05 ***0.01)
'*' found where number expected
r(7);

130 . restore

131 . export excel using "/Users/chenxin/Desktop/Processed data.xls"
file /Users/chenxin/Desktop/Processed data.xls saved

132 . label define njlab 1 "NJ" 0 "PA"

133 . label values nj njlab

134 .
135 . count
  410

136 . count if nj==1
  331

137 . count if nj==0
  79

138 .
139 . tab status2 nj, col
```

Key
frequency *column percentage*

STATUS2	nj PA	nj NJ	Total

0	0 0.00	1 0.30	1 0.24
1	78 98.73	321 96.98	399 97.32
2	0 0.00	2 0.60	2 0.49
3	1 1.27	5 1.51	6 1.46
4	0 0.00	1 0.30	1 0.24
5	0 0.00	1 0.30	1 0.24
Total	79 100.00	331 100.00	410 100.00

```
140 .
141 .
142 . count if status2 ==4
      1

143 . count if status2 ==2
      2

144 . count if status2 ==3
      6

145 . count if status2 ==. & nj~=.
      0

146 . count if status2 ==1
      399

147 . bysort nj: summ fte1 fte2 demp wage_st wage_st2 gap
```

-> nj = PA

Variable	Obs	Mean	Std. dev.	Min	Max
fte1	77	23.33117	11.85628	7.5	70.5
fte2	77	21.16558	8.276732	0	43.5
demp	75	-2.283333	10.85363	-41.5	22.75
wage_st	76	4.630132	.3516873	4.25	5.5
wage_st2	71	4.617465	.3574781	4.25	6.25
gap	79	0	0	0	0

```
-> nj = NJ
```

Variable	Obs	Mean	Std. dev.	Min	Max
fte1	321	20.43941	9.106239	5	85
fte2	320	20.96172	9.352609	0	60.5
demp	310	.4070968	8.503438	-34	34
wage_st	314	4.612134	.3463505	4.25	5.75
wage_st2	318	5.080849	.1045445	5	5.75
gap	331	.0995381	.0734713	0	.1882353

```
148 .
149 . * Example: test pre-policy FTE (fte1)

150 . ttest fte1, by(nj)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
PA	77	23.33117	1.351149	11.85628	20.64012	26.02222
NJ	321	20.43941	.5082607	9.106239	19.43945	21.43936
Combined	398	20.99887	.4887136	9.749805	20.03808	21.95966
diff		2.891761	1.230207		.4732069	5.310315

```
    diff = mean(PA) - mean(NJ)                                    t =   2.3506
H0: diff = 0                                     Degrees of freedom =      396

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.9904         Pr(|T| > |t|) = 0.0192          Pr(T > t) = 0.0096

151 .
152 . * post-policy FTE (fte2)

153 . ttest fte2, by(nj)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
PA	77	21.16558	.9432212	8.276732	19.287	23.04417
NJ	320	20.96172	.5228267	9.352609	19.93309	21.99034
Combined	397	21.00126	.4589353	9.144222	20.09901	21.90351
diff		.2038657	1.162128		-2.080863	2.488594

```
    diff = mean(PA) - mean(NJ)                                    t =   0.1754
H0: diff = 0                                     Degrees of freedom =      395
```

```
     Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
  Pr(T < t) = 0.5696         Pr(|T| > |t|) = 0.8608          Pr(T > t) = 0.4304
```

```
154 .
155 . * change in FTE employment demp = fte2-fte1

156 . ttest demp, by(nj)

Two-sample t test with equal variances
```

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
PA	75	-2.283333	1.253269	10.85363	-4.780526	.2138597
NJ	310	.4070968	.4829628	8.503438	-.5432151	1.357409
Combined	385	-.117013	.4615765	9.056784	-1.024547	.7905206
diff		-2.69043	1.158843		-4.96892	-.4119402

```
    diff = mean(PA) - mean(NJ)                                    t =  -2.3217
H0: diff = 0                     Degrees of freedom =      383

     Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
  Pr(T < t) = 0.0104         Pr(|T| > |t|) = 0.0208          Pr(T > t) = 0.9896
```

```
157 .
158 . * wave-1 starting wage

159 . ttest wage_st, by(nj)

Two-sample t test with equal variances
```

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
PA	76	4.630132	.0403413	.3516873	4.549768	4.710496
NJ	314	4.612134	.0195457	.3463505	4.573676	4.650591
Combined	390	4.615641	.0175718	.3470151	4.581093	4.650189
diff		.0179978	.0444095		-.0693156	.1053113

```
    diff = mean(PA) - mean(NJ)                                    t =   0.4053
H0: diff = 0                     Degrees of freedom =      388

     Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
  Pr(T < t) = 0.6572         Pr(|T| > |t|) = 0.6855          Pr(T > t) = 0.3428
```

```
160 .
161 . * wave-2 starting wage

162 . ttest wage_st2, by(nj)

Two-sample t test with equal variances
```

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
PA	**71**	**4.617465**	**.0424249**	**.3574781**	**4.532851**	**4.702078**
NJ	**318**	**5.080849**	**.0058626**	**.1045445**	**5.069315**	**5.092384**
Combined	**389**	**4.996273**	**.0128373**	**.2531905**	**4.971033**	**5.021512**
diff		**-.4633844**	**.0235051**		**-.5095982**	**-.4171707**

diff = mean(**PA**) - mean(**NJ**) t = **-19.7142**
H0: diff = 0 Degrees of freedom = **387**

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = **0.0000** Pr(|T| > |t|) = **0.0000** Pr(T > t) = **1.0000**

```
163 .
164 . * wage gap variable

165 . ttest gap, by(nj)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
PA	**79**	**0**	**0**	**0**	**0**	**0**
NJ	**331**	**.0995381**	**.0040383**	**.0734713**	**.0915939**	**.1074822**
Combined	**410**	**.0803588**	**.0037936**	**.0768139**	**.0729015**	**.0878161**
diff		**-.0995381**	**.0082739**		**-.1158028**	**-.0832733**

diff = mean(**PA**) - mean(**NJ**) t = **-12.0304**
H0: diff = 0 Degrees of freedom = **408**

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = **0.0000** Pr(|T| > |t|) = **0.0000** Pr(T > t) = **1.0000**

```
166 .
167 .
168 . twoway

169 . (hist wage_st if nj==1, percent width(0.1) start(4.25) color(black))
( is not a valid command name
r(199);

170 . (hist wage_st if nj==0, percent width(0.1) start(4.25) color(gs12)),
( is not a valid command name
r(199);

171 . title("February 1992") ytitle("Percent of Stores") ylabel(0(5)35) legend(order(1
>  "NJ" 2 "PA")) saving(feb2.gph, replace)
command title is unrecognized
r(199);
```

```
172 .
173 . twoway

174 . (hist wage_st2 if nj==1, percent width(0.1) start(4.25) color(black))
    ( is not a valid command name
    r(199);

175 . (hist wage_st2 if nj==0, percent width(0.1) start(4.25) color(gs12)),
    ( is not a valid command name
    r(199);

176 . title("November 1992") ytitle("Percent of Stores") ylabel(0(10)90) legend(off) s
    > aving(nov2.gph, replace)
    command title is unrecognized
    r(199);

177 .
178 . graph combine feb2.gph nov2.gph, col(1) xcommon
    file feb2.gph not found
    r(601);

179 .
180 . twoway

181 . (hist wage_st if nj==1, percent width(0.1) start(4.25) color(black))
    ( is not a valid command name
    r(199);

182 . (hist wage_st if nj==0, percent width(0.1) start(4.25) color(gs12)),
    ( is not a valid command name
    r(199);

183 . title("February 1992") ytitle("Percent of Stores") ylabel(0(5)35) legend(order(1
    >  "NJ" 2 "PA")) saving(feb2.gph, replace)
    command title is unrecognized
    r(199);

184 . twoway

185 . hist wage_st if nj==1, percent width(0.1) start(4.25) color(black)
    (bin=15, start=4.25, width=.1)

186 . hist wage_st if nj==0, percent width(0.1) start(4.25) color(gs12),
    (bin=13, start=4.25, width=.1)

187 . title("February 1992") ytitle("Percent of Stores") ylabel(0(5)35) legend(order(1
    >  "NJ" 2 "PA")) saving(feb2.gph, replace)
    command title is unrecognized
    r(199);

188 . * February (wage_st)

189 . twoway
```

```
190 . hist wage_st if nj==1, percent width(0.1) start(4.25) color(black))
    option ) not allowed
    r(198);

191 . hist wage_st if nj==0, percent width(0.1) start(4.25) color(gs12)),
    option ) not allowed
    r(198);

192 . title("February 1992") ytitle("Percent of Stores") ylabel(0(5)35) legend(order(1
    >  "NJ" 2 "PA")) saving(feb2.gph, replace)
    command title is unrecognized
    r(199);

193 .
194 . * November (wage_st2)

195 . twoway

196 . hist wage_st2 if nj==1, percent width(0.1) start(4.25) color(black))
    option ) not allowed
    r(198);

197 . hist wage_st2 if nj==0, percent width(0.1) start(4.25) color(gs12)),
    option ) not allowed
    r(198);

198 . title("November 1992") ytitle("Percent of Stores") ylabel(0(10)90) legend(off) s
    > aving(nov2.gph, replace)
    command title is unrecognized
    r(199);

199 .
200 . graph combine feb2.gph nov2.gph, col(1) xcommon
    file feb2.gph not found
    r(601);

201 .
202 . * February (wage_st)

203 . twoway

204 . hist wage_st if nj==1, percent width(0.1) start(4.25) color(black)
    (bin=15, start=4.25, width=.1)

205 . hist wage_st if nj==0, percent width(0.1) start(4.25) color(gs12),
    (bin=13, start=4.25, width=.1)

206 . title("February 1992") ytitle("Percent of Stores") ylabel(0(5)35) legend(order(1
    >  "NJ" 2 "PA")) saving(feb2.gph, replace)
    command title is unrecognized
    r(199);

207 .
```

```
208 . * November (wage_st2)

209 . twoway

210 . hist wage_st2 if nj==1, percent width(0.1) start(4.25) color(black)
    (bin=15, start=4.25, width=.1)

211 . hist wage_st2 if nj==0, percent width(0.1) start(4.25) color(gs12),
    (bin=20, start=4.25, width=.1)

212 . title("November 1992") ytitle("Percent of Stores") ylabel(0(10)90) legend(off) s
    > aving(nov2.gph, replace)
    command title is unrecognized
    r(199);

213 .
214 . graph combine feb2.gph nov2.gph, col(1) xcommon
    file feb2.gph not found
    r(601);

215 .
216 . hist wage_st2 if nj==1, percent width(0.1) start(4.25) color(black)
    (bin=15, start=4.25, width=.1)

217 . hist wage_st2 if nj==0, percent width(0.1) start(4.25) color(gs12),
    (bin=20, start=4.25, width=.1)

218 . hist wage_st2 if nj==1, percent width(0.1) start(4.25) color(black)
    (bin=15, start=4.25, width=.1)

219 . twoway

220 . hist wage_st if nj==0, percent width(0.1) start(4.25) color(gs12),
    (bin=13, start=4.25, width=.1)

221 . hist wage_st if nj==1, percent width(0.1) start(4.25) color(black)
    (bin=15, start=4.25, width=.1)

222 . title("November 1992") ytitle("Percent of Stores") ylabel(0(10)90) legend(off) s
    > aving(nov2.gph, replace)
    command title is unrecognized
    r(199);

223 . hist wage_st2 if nj==1, percent width(0.1) start(4.25) color(black))
    option ) not allowed
    r(198);

224 . hist wage_st if nj==1, percent width(0.1) start(4.25) color(black))
    option ) not allowed
    r(198);

225 . * 计算NJ整体demp加权均值

226 . summarize demp if nj==1
```

Variable	Obs	Mean	Std. dev.	Min	Max
demp	**310**	**.4070968**	**8.503438**	**-34**	**34**

```
227 . local mean_nj = r(mean)

228 . summarize demp if nj==0
```

Variable	Obs	Mean	Std. dev.	Min	Max
demp	**75**	**-2.283333**	**10.85363**	**-41.5**	**22.75**

```
229 . local mean_pa = r(mean)

230 . local did = `mean_nj' - `mean_pa'

231 . display "NJ平均就业变化: " `mean_nj'
NJ平均就业变化: .40709678

232 . display "PA平均就业变化: " `mean_pa'
PA平均就业变化: -2.2833333

233 . display "简单DiD = " `did'
简单DiD = 2.6904301

234 .
235 .
236 . replace gap = 0 if missing(gap)
(0 real changes made)

237 . gen frac_full  = empft / (empft + emppt)
(6 missing values generated)

238 . gen frac_full2 = empft2 / (empft2 + emppt2)
(19 missing values generated)

239 . gen d_fracfull = frac_full2 - frac_full
(25 missing values generated)

240 . gen d_hrsopen = hrsopen2 - hrsopen
(11 missing values generated)

241 . gen d_nregs11 = nregs112 - nregs11
(38 missing values generated)

242 . gen d_meals = meals2 - meals
(11 missing values generated)

243 . gen d_inctime = inctime2 - inctime
(89 missing values generated)

244 . gen d_firstinc = firstin2 - firstinc
(114 missing values generated)
```

```
245 . gen slope  = firstinc / inctime
    (61 missing values generated)

246 . gen slope2 = firstin2 / inctime2
    (92 missing values generated)

247 . gen d_slope = slope2 - slope
    (133 missing values generated)

248 . global chain_ctrl   bk kfc rr

249 . global firm_ctrl    co_owned

250 . global region_ctrl  southj centralj northj pa2

251 . capture ssc install eststo

252 . eststo coliv_1: reg d_fracfull nj $chain_ctrl $firm_ctrl, robust
    variable rr not found
    r(111);

253 . eststo colv_1:  reg d_fracfull gap $chain_ctrl $firm_ctrl, robust
    variable rr not found
    r(111);

254 . eststo colvi_1: reg d_fracfull gap $chain_ctrl $firm_ctrl $region_ctrl, robust
    variable rr not found
    r(111);

255 . reg d_fracfull gap chain co_owned southj centralj northj, robust

    Linear regression                               Number of obs     =        385
                                                    F(6, 378)         =       2.38
                                                    Prob > F          =     0.0286
                                                    R-squared         =     0.0321
                                                    Root MSE          =     .31046

------------------------------------------------------------------------------
             |               Robust
  d_fracfull | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         gap |   .1907448   .2452643     0.78   0.437    -.2915084    .6729981
       chain |  -.0251864   .0149444    -1.69   0.093    -.0545709    .0041981
    co_owned |   .0213781   .0327728     0.65   0.515    -.0430617    .0858179
      southj |   .0481455   .0557956     0.86   0.389    -.0615631    .1578541
    centralj |  -.0103325   .0555696    -0.19   0.853    -.1195967    .0989318
      northj |   .0890768    .052386     1.70   0.090    -.0139277    .1920813
       _cons |  -.0010167   .0520082    -0.02   0.984    -.1032784    .1012449
------------------------------------------------------------------------------

256 . estimates store model1

257 . reg d_hrsopen gap chain co_owned southj centralj northj, robust
```

```
Linear regression                               Number of obs     =        399
                                                F(6, 392)         =       1.69
                                                Prob > F          =     0.1212
                                                R-squared         =     0.0312
                                                Root MSE          =     .98696

------------------------------------------------------------------------------
             |               Robust
   d_hrsopen |   Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         gap |   .0521619   .7211344     0.07   0.942    -1.365613    1.469937
       chain |  -.0924265   .0394987    -2.34   0.020    -.1700823   -.0147708
   co_owned |  -.1887828   .1338223    -1.41   0.159     -.451882    .0743164
      southj |  -.0155949   .1461122    -0.11   0.915    -.3028565    .2716667
    centralj |  -.3151245   .2069226    -1.52   0.129    -.7219414    .0916924
      northj |  -.1112363   .1150102    -0.97   0.334    -.3373503    .1148777
       _cons |   .3746301   .1292339     2.90   0.004     .1205519    .6287083
------------------------------------------------------------------------------
```

```
258 . estimates store model2

259 . reg d_nregs11 gap chain co_owned southj centralj northj, robust

Linear regression                               Number of obs     =        372
                                                F(6, 365)         =       1.06
                                                Prob > F          =     0.3879
                                                R-squared         =     0.0122
                                                Root MSE          =     .90985

------------------------------------------------------------------------------
             |               Robust
   d_nregs11 |   Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         gap |    -.42588   .7834497    -0.54   0.587    -1.966522    1.114762
       chain |   .0427411   .0390015     1.10   0.274    -.0339548     .119437
   co_owned |  -.0718576   .1082347    -0.66   0.507    -.2846994    .1409842
      southj |   .2811587   .1451038     1.94   0.053    -.0041857    .5665031
    centralj |   .1368085   .1943145     0.70   0.482     -.245308     .518925
      northj |   .2145084   .1405248     1.53   0.128    -.0618315    .4908482
       _cons |  -.2828046   .1255033    -2.25   0.025    -.5296049   -.0360043
------------------------------------------------------------------------------
```

```
260 . estimates store model2

261 . reg d_meals gap chain co_owned southj centralj northj, robust

Linear regression                               Number of obs     =        399
                                                F(6, 392)         =       1.11
                                                Prob > F          =     0.3536
                                                R-squared         =     0.0195
                                                Root MSE          =     .58854

-------------+
```

d_meals	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	-.9613206	.4939158	-1.95	0.052	-1.932376	.0097348
chain	-.0113225	.0270397	-0.42	0.676	-.0644835	.0418384
co_owned	-.0512012	.06146	-0.83	0.405	-.1720336	.0696312
southj	.0147654	.0987424	0.15	0.881	-.1793656	.2088964
centralj	.1475373	.1134672	1.30	0.194	-.0755431	.3706177
northj	.0869222	.0759712	1.14	0.253	-.0624398	.2362843
_cons	-.0727628	.0676813	-1.08	0.283	-.2058266	.060301

```
262 . estimates store model2

263 . reg d_inctime gap chain co_owned southj centralj northj, robust

Linear regression                               Number of obs     =        321
                                                F(6, 314)         =       1.18
                                                Prob > F          =     0.3176
                                                R-squared         =     0.0265
                                                Root MSE          =     14.478
```

d_inctime	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	-3.765247	12.64455	-0.30	0.766	-28.64401	21.11352
chain	1.086338	.7902513	1.37	0.170	-.4685195	2.641195
co_owned	-3.020086	1.844037	-1.64	0.102	-6.648317	.6081439
southj	5.262978	2.873547	1.83	0.068	-.3908621	10.91682
centralj	3.180241	3.243239	0.98	0.328	-3.200987	9.56147
northj	1.599885	2.617874	0.61	0.542	-3.550907	6.750677
_cons	.0404631	2.738231	0.01	0.988	-5.347137	5.428063

```
264 . estimates store model2

265 . reg d_firstinc gap chain co_owned southj centralj northj, robust

Linear regression                               Number of obs     =        296
                                                F(6, 289)         =       0.30
                                                Prob > F          =     0.9345
                                                R-squared         =     0.0071
                                                Root MSE          =     .14141
```

d_firstinc	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	.0214738	.1377722	0.16	0.876	-.2496904	.2926379
chain	-.0078647	.007398	-1.06	0.289	-.0224256	.0066961
co_owned	.0146255	.0197281	0.74	0.459	-.0242033	.0534544
southj	.0050856	.0257793	0.20	0.844	-.0456533	.0558245
centralj	.020076	.0343269	0.58	0.559	-.0474865	.0876385

northj	.0005549	.023999	0.02	0.982	-.0466802	.0477899
_cons	-.0060024	.0246715	-0.24	0.808	-.0545609	.0425561

```
266 . estimates store model2

267 . reg d_slope gap chain co_owned southj centralj northj, robust

Linear regression                               Number of obs     =        277
                                                F(6, 270)         =       1.04
                                                Prob > F          =     0.3989
                                                R-squared         =     0.0238
                                                Root MSE          =     .02728
```

d_slope	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	.0132066	.023016	0.57	0.567	-.0321071	.0585203
chain	-.0009929	.0016011	-0.62	0.536	-.0041452	.0021594
co_owned	.0069305	.004247	1.63	0.104	-.001431	.0152919
southj	-.0030032	.0055764	-0.54	0.591	-.0139819	.0079754
centralj	.0079617	.00762	1.04	0.297	-.0070404	.0229638
northj	.0000219	.005021	0.00	0.997	-.0098634	.0099072
_cons	-.0052442	.0052505	-1.00	0.319	-.0155813	.0050929

```
268 . estimates store model2

269 . reg d_fracfull gap bk kfc roys co_owned southj centralj northj, robust

Linear regression                               Number of obs     =        385
                                                F(8, 376)         =       2.02
                                                Prob > F          =     0.0436
                                                R-squared         =     0.0357
                                                Root MSE          =     .31071
```

d_fracfull	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	.1841125	.2419215	0.76	0.447	-.2915761	.6598011
bk	.0589818	.0512412	1.15	0.250	-.0417734	.1597369
kfc	.0676225	.0606163	1.12	0.265	-.051567	.1868119
roys	-.0120005	.0527404	-0.23	0.820	-.1157036	.0917027
co_owned	.027661	.0375024	0.74	0.461	-.0460797	.1014017
southj	.0440536	.0562275	0.78	0.434	-.0665062	.1546134
centralj	-.0084245	.0563571	-0.15	0.881	-.1192392	.1023901
northj	.0907354	.0524347	1.73	0.084	-.0123665	.1938374
_cons	-.091252	.0558109	-1.64	0.103	-.2009926	.0184886

```
270 . reg d_hrsopen gap bk kfc roys co_owned southj centralj northj, robust
```

Linear regression | Number of obs = 399
F(8, 390) = 2.57
Prob > F = 0.0097
R-squared = 0.0755
Root MSE = .9666

d_hrsopen	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	.0361901	.7114986	0.05	0.959	-1.362663	1.435043
bk	.0689986	.0983955	0.70	0.484	-.1244535	.2624507
kfc	.2819078	.1298392	2.17	0.031	.0266356	.5371801
roys	-.4189691	.1829183	-2.29	0.023	-.7785985	-.0593398
co_owned	-.0818281	.1323673	-0.62	0.537	-.3420708	.1784146
southj	-.0490435	.1395635	-0.35	0.725	-.3234345	.2253475
centralj	-.2797685	.1988514	-1.41	0.160	-.6707233	.1111864
northj	-.0874924	.1069636	-0.82	0.414	-.2977897	.1228049
_cons	.1499596	.1076617	1.39	0.164	-.0617104	.3616295

271 . reg d_nregs11 gap bk kfc roys co_owned southj centralj northj, robust

Linear regression | Number of obs = 372
F(8, 363) = 1.53
Prob > F = 0.1449
R-squared = 0.0227
Root MSE = .9075

d_nregs11	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	-.4438801	.7877978	-0.56	0.573	-1.993101	1.105341
bk	-.1994706	.0984033	-2.03	0.043	-.3929827	-.0059584
kfc	.0320291	.1200293	0.27	0.790	-.204011	.2680693
roys	-.1991706	.1617209	-1.23	0.219	-.5171981	.1188569
co_owned	-.0533649	.1256983	-0.42	0.671	-.3005532	.1938234
southj	.2675359	.1436541	1.86	0.063	-.0149629	.5500348
centralj	.1448463	.1934131	0.75	0.454	-.2355045	.5251971
northj	.2253759	.1390165	1.62	0.106	-.0480029	.4987548
_cons	-.0760263	.1000217	-0.76	0.448	-.2727211	.1206685

272 . reg d_meals gap bk kfc roys co_owned southj centralj northj, robust

Linear regression | Number of obs = 399
F(8, 390) = 3.98
Prob > F = 0.0002
R-squared = 0.0801
Root MSE = .57154

	Robust

d_meals	Coefficient	std. err.	t	P>|t|	[95% conf.	interval]
gap	-.8697999	.4847936	-1.79	0.074	-1.822936	.083336
bk	.1017653	.0881662	1.15	0.249	-.0715752	.2751058
kfc	-.2613684	.1097154	-2.38	0.018	-.4770759	-.0456608
roys	.131677	.0835165	1.58	0.116	-.0325219	.2958758
co_owned	-.0447402	.0648002	-0.69	0.490	-.1721417	.0826613
southj	.0531903	.0968772	0.55	0.583	-.1372766	.2436571
centralj	.1548728	.1026133	1.51	0.132	-.0468715	.3566172
northj	.0749548	.0729971	1.03	0.305	-.0685623	.2184719
_cons	-.1317881	.0889574	-1.48	0.139	-.3066842	.043108

273 . reg d_inctime gap bk kfc roys co_owned southj centralj northj, robust

Linear regression

Number of obs	=	321
F(8, 312)	=	1.26
Prob > F	=	0.2661
R-squared	=	0.0394
Root MSE	=	14.427

d_inctime	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	-5.087166	12.58832	-0.40	0.686	-29.8559	19.68157
bk	-1.180482	2.469641	-0.48	0.633	-6.039739	3.678774
kfc	.8245158	2.746699	0.30	0.764	-4.57988	6.228912
roys	4.588316	2.724574	1.68	0.093	-.7725463	9.949179
co_owned	-4.730356	2.138343	-2.21	0.028	-8.937752	-.522959
southj	5.172015	2.86626	1.80	0.072	-.4676283	10.81166
centralj	2.568081	3.240734	0.79	0.429	-3.808376	8.944538
northj	1.470709	2.592016	0.57	0.571	-3.629333	6.57075
_cons	2.423545	2.698699	0.90	0.370	-2.886405	7.733495

274 . reg d_firstinc gap bk kfc roys co_owned southj centralj northj, robust

Linear regression

Number of obs	=	296
F(8, 287)	=	1.46
Prob > F	=	0.1705
R-squared	=	0.0376
Root MSE	=	.13971

d_firstinc	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	.0285424	.1373849	0.21	0.836	-.2418674	.2989522
bk	-.0008707	.0219759	-0.04	0.968	-.0441251	.0423837
kfc	.031654	.0281601	1.12	0.262	-.0237725	.0870804
roys	-.047032	.0233592	-2.01	0.045	-.0930091	-.0010549
co_owned	.0233054	.0224421	1.04	0.300	-.0208666	.0674774
southj	-.0020399	.0257971	-0.08	0.937	-.0528154	.0487355

centralj	.0170786	.0359252	0.48	0.635	-.0536317	.087789
northj	.0003219	.0231798	0.01	0.989	-.045302	.0459459
_cons	-.019045	.0216752	-0.88	0.380	-.0617076	.0236176

```
275 . reg d_slope gap bk kfc roys co_owned southj centralj northj, robust

Linear regression                               Number of obs     =        277
                                                F(8, 268)         =       0.98
                                                Prob > F          =     0.4483
                                                R-squared         =     0.0328
                                                Root MSE          =     .02725
```

d_slope	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gap	.0149211	.0227285	0.66	0.512	-.029828	.0596702
bk	-.0003666	.0047903	-0.08	0.939	-.0097979	.0090648
kfc	-.0009693	.0057751	-0.17	0.867	-.0123396	.0104011
roys	-.0076104	.004521	-1.68	0.093	-.0165117	.0012909
co_owned	.0091759	.0046111	1.99	0.048	.0000973	.0182544
southj	-.0032689	.005698	-0.57	0.567	-.0144874	.0079496
centralj	.0084968	.0078024	1.09	0.277	-.006865	.0238585
northj	.0000321	.0050069	0.01	0.995	-.0098257	.0098898
_cons	-.0061083	.0054629	-1.12	0.265	-.0168639	.0046472

```
276 . gen gap_coowned = gap * co_owned

277 . gen d_fte = fte2 - fte
    fte ambiguous abbreviation
    r(111);

278 . gen d_fte = fte2 - fte1
    (25 missing values generated)

279 . gen gap = 0
    variable gap already defined
    r(110);

280 . replace gap = (5.05 - wage_st)/wage_st if state==1 & wage_st <5.05
    (0 real changes made)

281 . gen gap_coowned = gap * co_owned
    variable gap_coowned already defined
    r(110);

282 . drop gap_coowned

283 . gen gap_coowned = gap * co_owned

284 .  gen gap1 = 0
```

```
285 .  replace gap1 = (5.05 - wage_st)/wage_st if state==1 & wage_st <5.05
(286 real changes made)

286 . drop gap_coowned

287 . gen gap1_coowned = gap1 * co_owned

288 . reg d_fte gap1 co_owned gap1_coowned bk kfc roys southj centralj northj, robust

Linear regression                               Number of obs     =        385
                                                F(9, 375)         =       1.73
                                                Prob > F          =     0.0811
                                                R-squared         =     0.0365
                                                Root MSE          =     8.9958

------------------------------------------------------------------------------
             |               Robust
       d_fte |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        gap1 |   11.46165   8.779784     1.31   0.193    -5.802131    28.72543
    co_owned |   1.354693   1.392894     0.97   0.331    -1.384169    4.093554
gap1_coowned |   -14.0589   11.48092    -1.22   0.222    -36.63395    8.516144
          bk |   .2932062    1.73498     0.17   0.866    -3.118302    3.704714
         kfc |   .5448369   1.646716     0.33   0.741    -2.693117     3.78279
        roys |  -2.053224    1.71697    -1.20   0.233    -5.429319    1.322871
      southj |   2.217109   1.547524     1.43   0.153    -.8258026    5.260021
    centralj |   .8343841   1.921318     0.43   0.664    -2.943524    4.612292
      northj |   2.350934   1.630105     1.44   0.150    -.8543579    5.556226
       _cons |  -2.503747   1.876333    -1.33   0.183    -6.193199    1.185704
------------------------------------------------------------------------------

289 . reg d_fte gap1 bk kfc roys southj centralj northj if co_owned==1, robust

Linear regression                               Number of obs     =        134
                                                F(7, 126)         =       1.96
                                                Prob > F          =     0.0651
                                                R-squared         =     0.1370
                                                Root MSE          =     7.0386

------------------------------------------------------------------------------
             |               Robust
       d_fte |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        gap1 |  -3.282084   9.596094    -0.34   0.733    -22.27247     15.7083
          bk |   5.778364   2.872038     2.01   0.046     .0946852    11.46204
         kfc |   5.219962   2.733653     1.91   0.058     -.189857    10.62978
        roys |    1.00931   2.554449     0.40   0.693    -4.045871     6.06449
      southj |  -.0443005   1.696537    -0.03   0.979    -3.401698    3.313097
    centralj |  -1.901323   2.899987    -0.66   0.513    -7.640312    3.837666
      northj |   2.979332   1.818328     1.64   0.104    -.6190862     6.57775
       _cons |  -4.120054   2.827778    -1.46   0.148    -9.716144    1.476035
------------------------------------------------------------------------------

290 . reg d_fte gap1 bk kfc roys southj centralj northj if co_owned==0, robust
```

```
Linear regression                               Number of obs     =        251
                                                F(7, 243)         =       1.18
                                                Prob > F          =     0.3150
                                                R-squared         =     0.0302
                                                Root MSE          =      9.837

------------------------------------------------------------------------------
             |               Robust
       d_fte |  Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        gap1 |    13.06573   9.402535     1.39   0.166    -5.455138    31.58661
          bk |   -.5641379   1.999076    -0.28   0.778    -4.501866     3.37359
         kfc |    -.764501    1.98418    -0.39   0.700    -4.672889    3.143887
        roys |   -1.576914   2.240465    -0.70   0.482    -5.990124    2.836296
      southj |    2.958806   2.249396     1.32   0.190    -1.471997    7.389609
    centralj |    .9182112   2.530214     0.36   0.717    -4.065739    5.902161
      northj |    1.503425   2.282871     0.66   0.511    -2.993315    6.000165
       _cons |   -1.804916    2.27736    -0.79   0.429      -6.2908    2.680969
------------------------------------------------------------------------------
```

```
291 . reg dln_pmeal gap1 co_owned gap1_coowned bk kfc roys southj centralj northj, robust

Linear regression                               Number of obs     =        356
                                                F(9, 346)         =       2.61
                                                Prob > F          =     0.0062
                                                R-squared         =     0.0815
                                                Root MSE          =     .09975

------------------------------------------------------------------------------
             |               Robust
   dln_pmeal |  Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        gap1 |   -.0064901   .0853147    -0.08   0.939    -.1742908    .1613105
    co_owned |   -.0210489   .0186003    -1.13   0.259    -.0576328    .0155351
gap1_coowned |    .2879765   .1344998     2.14   0.033     .0234364    .5525165
          bk |   -.0680934   .0255391    -2.67   0.008    -.1183249    -.017862
         kfc |   -.0683566    .026957    -2.54   0.012    -.1213768   -.0153363
        roys |   -.0684443     .03007    -2.28   0.023    -.1275873   -.0093012
      southj |    .0201713   .0187653     1.07   0.283    -.0167372    .0570798
    centralj |    .0045121   .0229162     0.20   0.844    -.0405605    .0495847
      northj |    .0293689   .0162882     1.80   0.072    -.0026676    .0614053
       _cons |     .055473   .0269654     2.06   0.040     .0024364    .1085097
------------------------------------------------------------------------------
```

```
292 .
293 . export excel using "/Users/chenxin/Desktop/final data.xls"
    file /Users/chenxin/Desktop/final data.xls saved

294 .
```